



DR ESLAM AHMED
eliteigcse.com

Volume and Surface Area

Prisms: area of the cross-section $\times$ length. Cones and pyramids: one third of that. Surface area: add the area of every face.

THE TOPIC ON ONE MAP

Volume and Surface Area

01 — Prisms and cylinders

cross-section $\times$ length $\pi r^2 h$

02 — Cones, pyramids and spheres

$\frac{1}{3}\pi r^2 h$ $\frac{4}{3}\pi r^3$ frustum

03 — Surface area

net of a prism $\pi r l, 4\pi r^2$

04 — Problem solving

fill / empty at a rate melt and recast

FORMULA SHEET Volumes and curved surfaces you must recognise:

solid	volume	surface
prism	area $\times$ length	add every face
cylinder	$\pi r^2 h$	curved $2\pi r h$ (+ two circles $2\pi r^2$)
cone	$\frac{1}{3}\pi r^2 h$	curved $\pi r l$ (+ base πr^2)
pyramid	$\frac{1}{3} \times \text{base} \times h$	add the triangles + base
sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$
hemisphere	$\frac{2}{3}\pi r^3$	curved $2\pi r^2$ (+ flat πr^2)

01 Prisms and Cylinders

Find the area of the cross-section (the face that repeats), then multiply by the length. A cylinder is a prism with a circle.

WHEN YOU SEE → YOU DO

an L-shaped or trapezium prism

→ area of the end face first

“cylinder, radius r , height h ”

→ $\pi r^2 h$

“change 1 m^3 to cm^3 ”

→ $\times 100^3 = 1\,000\,000$

“volume in litres”

→ 1 litre = 1000 cm^3

Cross-section area

× length

Units

WORKED MODEL • JANUARY 2020 P2HR

Trapezium prism: $BC = 7$, $AD = 12$, length 10, volume 608. Find the height h .

$$1 \quad \frac{1}{2}(7 + 12)h \times 10 = 608$$

area × length

ANSWER $h = 6.4 \text{ cm}$

IDEA 01

Practise: Prisms and Cylinders

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Prism

CH 4 • TOPIC 7 2 marks

A prism has a cross-section of area 18 cm^2 and length 12 cm. Work out its volume.

Q2 DR ESLAM Cylinder

CH 4 • TOPIC 7 2 marks

A cylinder has radius 5 cm and height 8 cm. Work out its volume in terms of π .

SELECTED PRACTICE

Q3**PRACTICE**

Practice • Easy

CH 4 • TOPIC 7 3 marks

- 1 The diagram shows a prism.

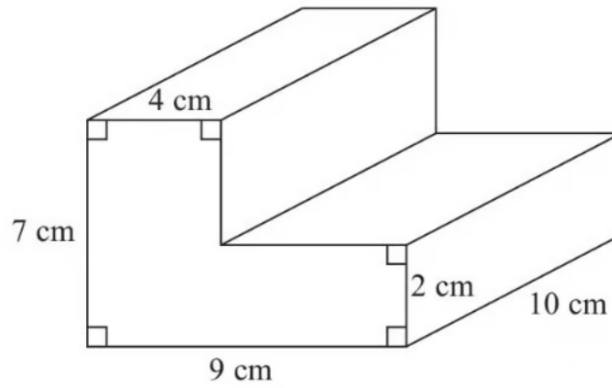


Diagram **NOT**
accurately drawn

Work out the volume of the prism.

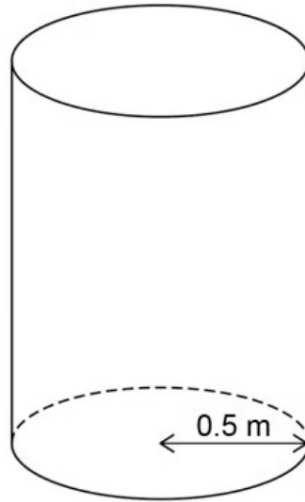
(3 marks)

Q4**PRACTICE**

Practice · Easy

CH 4 · TOPIC 7 2 marks

- 11** 3.8 m^3 of concrete is made into the shape of a cylinder.
The base has radius 0.5 metres.



Work out the height of the cylinder.

.....m

(2 marks)

Q5**PRACTICE**

Practice · Medium

CH 4 · TOPIC 7 3 marks

7 Jane makes cheese. The cheese is in the shape of a cuboid.

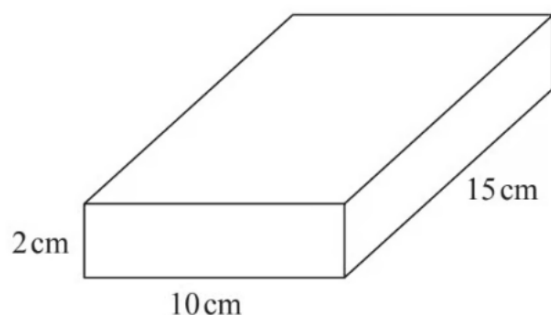


Diagram **NOT**
accurately drawn

Jane is going to make a new cheese.

The new cheese will also be in the shape of a cuboid.

The cross section of the cuboid will be a 5 cm by 5 cm square.

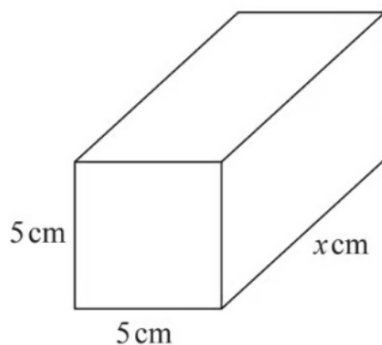


Diagram **NOT**
accurately drawn

Jane wants the new cuboid to have the same volume as the 2 cm by 10 cm by 15 cm cuboid.

Work out the value of x .

(3 marks)

Q6

PRACTICE

Practice · Very Hard

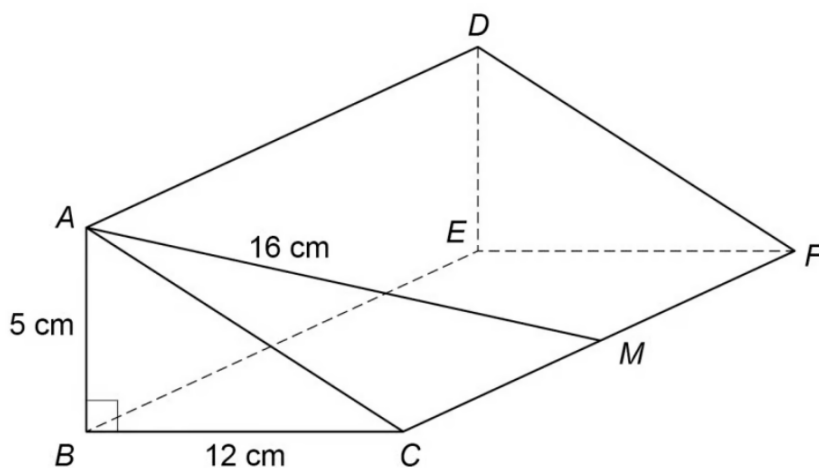
CH 4 · TOPIC 7 4 marks

13 Right-angled triangle ABC is the cross section of a prism.

$$AB = 5 \text{ cm} \quad BC = 12 \text{ cm}$$

M is the midpoint of CF .

$$AM = 16 \text{ cm}$$



Work out the volume of the prism.

.....cm³
(4 marks)

PAST PAPERS

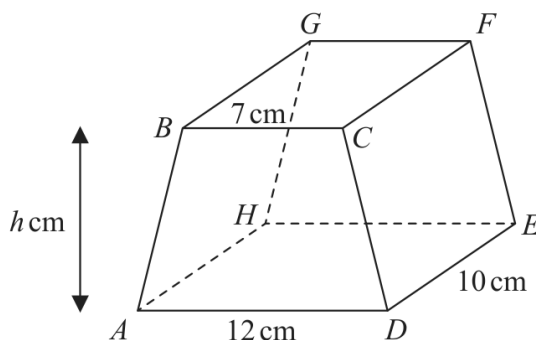
Q7

PAST PAPER

Edexcel 4MA1 · January 2020 · Paper 2HR · Q13

CH 4 · TOPIC 7 3 marks

13

Diagram **NOT**
accurately drawn

The diagram shows a prism $ABCDEFGH$ in which $ABCD$ is a trapezium with BC parallel to AD and $CDEF$ is a rectangle.

$$BC = 7 \text{ cm} \quad AD = 12 \text{ cm} \quad DE = 10 \text{ cm}$$

The height of trapezium $ABCD$ is $h \text{ cm}$

The volume of the prism is 608 cm^3

Work out the value of h .

$$h = \dots\dots\dots$$

(Total for Question 13 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q5

CH 4 • TOPIC 7 4 marks

- 5 The diagram shows a cuboid and a cylinder.

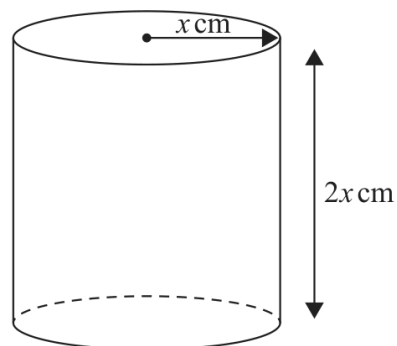
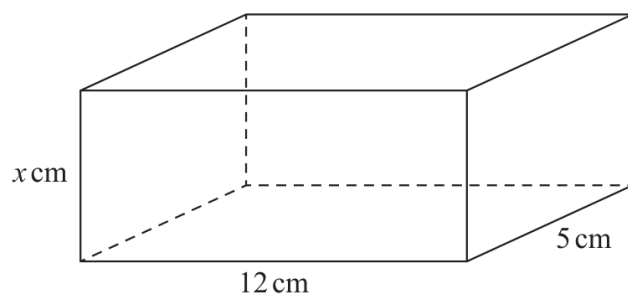


Diagram **NOT**
accurately drawn

The dimensions of the cuboid are $x\text{ cm}$ by 12 cm by 5 cm .
The volume of the cuboid is 270 cm^3

The radius of the cylinder is $x\text{ cm}$.
The height of the cylinder is $2x\text{ cm}$.

- (a) Work out the volume of the cylinder.
Give your answer correct to the nearest whole number.

..... cm^3

- (b) Change 1 m^3 to cm^3

..... cm^3

(Total for Question 5 is 4 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 2H · Q10

CH 4 · TOPIC 7 5 marks

10 Here is a triangular prism.

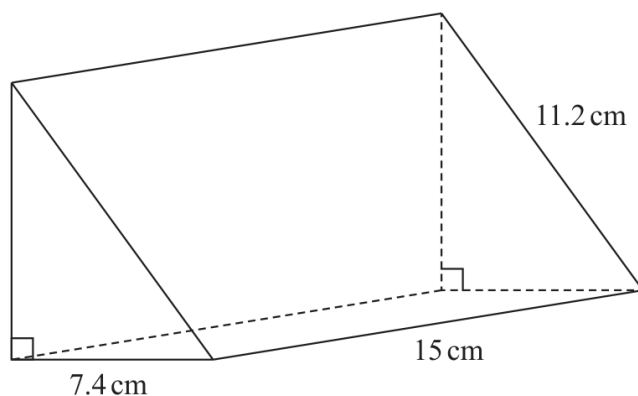


Diagram **NOT**
accurately drawn

Work out the volume of the prism.
Give your answer correct to 3 significant figures.

..... cm³

(Total for Question 10 is 5 marks)

Q10**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2HR · Q9

CH 4 · TOPIC 7 6 marks

- 9 The diagram shows a rectangular sheet of metal $ABCD$

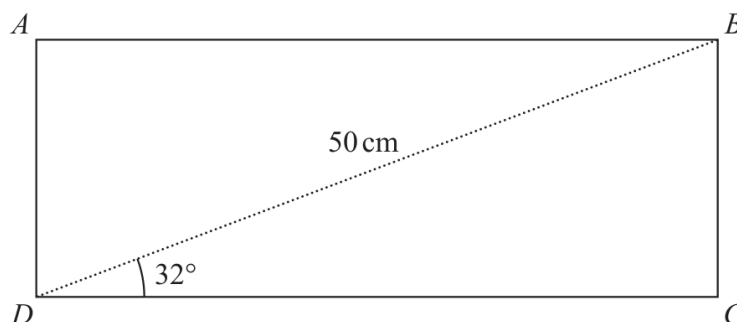


Diagram **NOT**
accurately drawn

$BD = 50$ cm and angle $BDC = 32^\circ$

Nasser joins side AD to side BC to form a cylinder.

BC is the height of the cylinder.

DC is the circumference of the cross section of the cylinder.

Work out the volume, in cm^3 , of the cylinder.

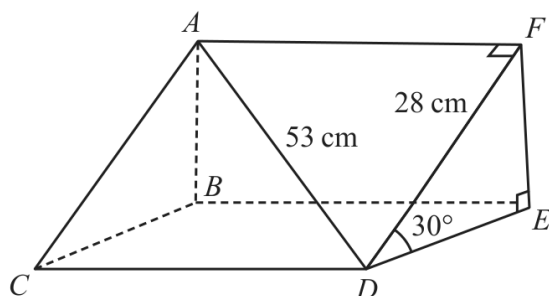
Give your answer correct to 3 significant figures.

..... cm^3

(Total for Question 9 is 6 marks)

Q11 PAST PAPER Edexcel 4MA1 • November 2023 • Paper 1H • Q21

CH 4 • TOPIC 7 5 marks

21 Here is a triangular prism $ABCDEF$ Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

(Total for Question 21 is 5 marks)

Q12**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q5

CH 4 · TOPIC 7 4 marks

- 5 The diagram shows a cylinder.

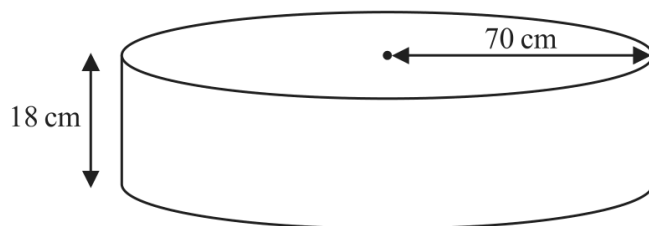


Diagram **NOT**
accurately drawn

The radius of the cylinder is 70 cm

The height of the cylinder is 18 cm

Work out the volume of the cylinder.

Give your answer in litres correct to the nearest litre.

..... litres

(Total for Question 5 is 4 marks)

Q13 PAST PAPER Edexcel 4MA1 · June 2025 · Paper 1H · Q7

CH 4 · TOPIC 7 3 marks

7 Here is a solid cylinder.

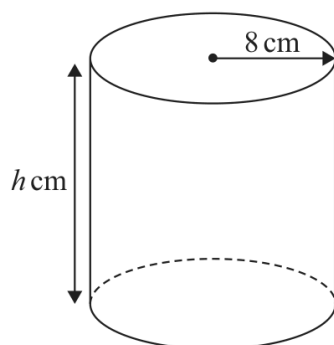


Diagram **NOT**
accurately drawn

The radius of the cylinder is 8 cm

The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h

Give your answer correct to one decimal place.

$h = \dots\dots\dots$

(Total for Question 7 is 3 marks)

02 Cones, Pyramids and Spheres

A cone or pyramid is one third of the prism around it. A frustum is a big cone minus a small cone (similar shapes).

WHEN YOU SEE → YOU DO

“cone” / “pyramid”

→ $\frac{1}{3} \times \text{base} \times \text{height}$

“sphere” / “hemisphere”

→ $\frac{4}{3}\pi r^3$ / $\frac{2}{3}\pi r^3$

“frustum”

→ large cone – small cone

“volume is $k\pi$, find k ”

→ keep π , add the parts

slant height given, height wanted

→ Pythagoras: $l^2 = r^2 + h^2$

WORKED MODEL · NOVEMBER 2024 P1H

A hemispherical bowl: outer radius 12 cm, inner radius 9 cm. Volume of metal?

1 $\frac{2}{3}\pi(12^3 - 9^3)$

outer – inner

ANSWER $666\pi = 2092 \text{ cm}^3$

IDEA 02

Practise: Cones, Pyramids and Spheres

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Cone

CH 4 • TOPIC 7 2 marks

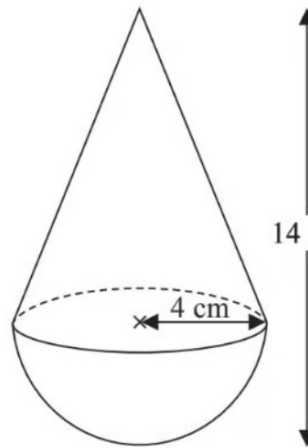
A cone has radius 6 cm and height 10 cm. Work out its volume to 3 s.f.

SELECTED PRACTICE

Q15 **PRACTICE** Practice · Medium

CH 4 · TOPIC 7 3 marks

- 6 The diagram shows a solid made from a hemisphere and a cone.



The radius of the hemisphere is 4 cm.

The radius of the base of the cone is 4 cm.

Calculate the volume of the solid.

Give your answer correct to 3 significant figures.

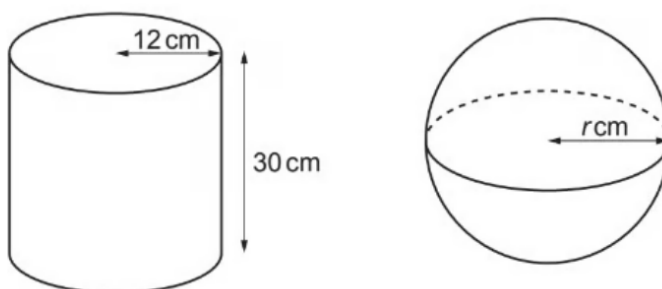
(3 marks)

Q16**PRACTICE**

Practice · Hard

CH 4 · TOPIC 7 **5 marks**

- 15** The diagram shows a cylinder and a sphere.



The cylinder has radius 12cm and height 30cm.

The cylinder and the sphere have the same volume.

Work out the radius r cm of the sphere.

..... cm

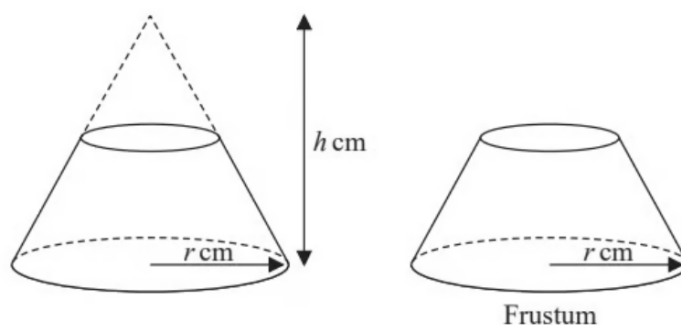
(5 marks)

Q17**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 7 4 marks

- 7 A frustum is made by removing a small cone from a large cone. The cones are mathematically similar.

Diagram **NOT**
accurately drawn

The large cone has base radius $r \text{ cm}$ and height $h \text{ cm}$. Given that

$$\frac{\text{volume of frustum}}{\text{volume of large cone}} = \frac{98}{125}$$

find an expression, in terms of h , for the height of the frustum.

..... cm

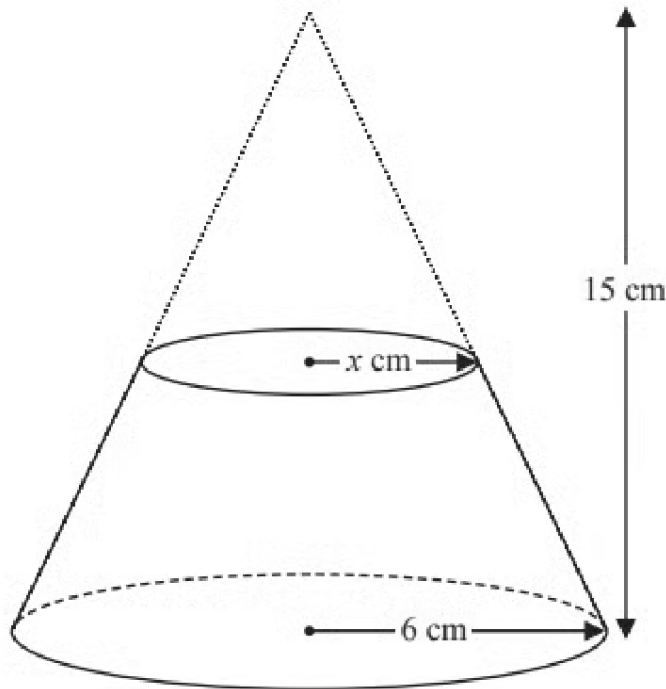
(4 marks)

Q18 PRACTICE Practice · Very Hard

CH 4 · TOPIC 7 5 marks

21 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.



The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm. The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25} \pi \text{ cm}^3$, work out the value of x

Show clear algebraic working.

(5 marks)

PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q26

CH 4 · TOPIC 7 6 marks

26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

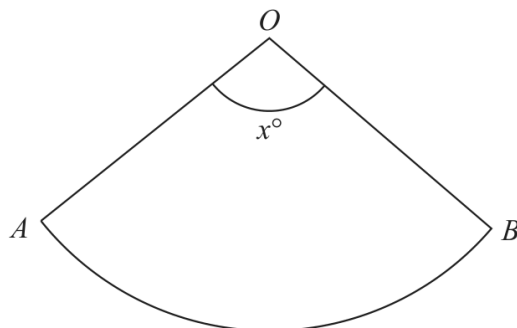


Diagram **NOT**
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

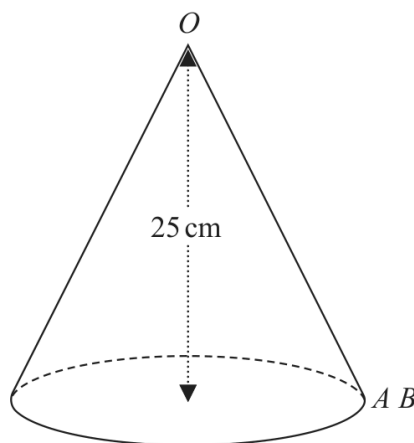


Diagram **NOT**
accurately drawn

The height of the cone is 25 cm.

The volume of the cone is 1600 cm^3

Work out the value of x .

Give your answer correct to the nearest whole number.

$x = \dots\dots\dots$

(Total for Question 26 is 6 marks)

Q20 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1H • Q22

CH 4 • TOPIC 7 5 marks

22 A solid is made from a cone and a hemisphere.

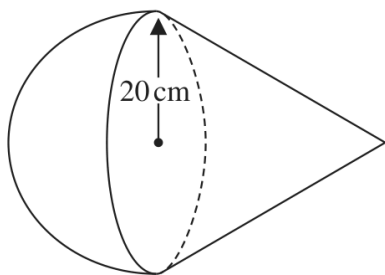


Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi\text{ cm}^2$

The volume of the solid is $k\pi\text{ cm}^3$

Work out the exact value of k

$k = \dots\dots\dots$

(Total for Question 22 is 5 marks)

Q21**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2HR · Q24

CH 4 · TOPIC 7 5 marks

24 The surface area of sphere **A** is nine times the surface area of sphere **B**

The difference between the volume of sphere **A** and the volume of sphere **B** is $117\pi \text{ cm}^3$

Find the radius of the smaller sphere.

Show your working clearly.

..... cm

(Total for Question 24 is 5 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1H · Q23

CH 4 · TOPIC 7 5 marks

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

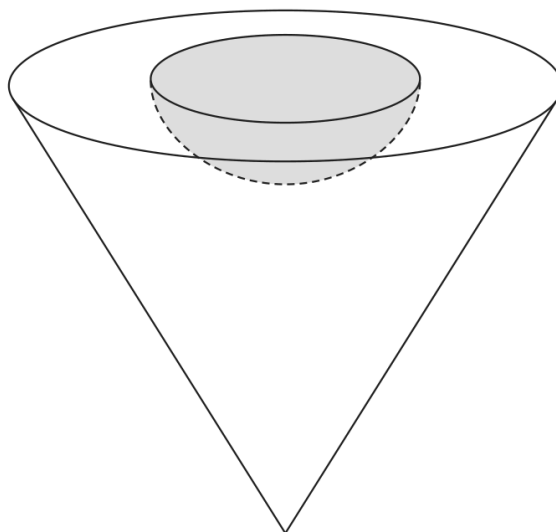


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm

The radius of the base of the cone is $5x$ cm

The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

(Total for Question 23 is 5 marks)

Q23**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 1H · Q22

CH 4 · TOPIC 7 3 marks

- 22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

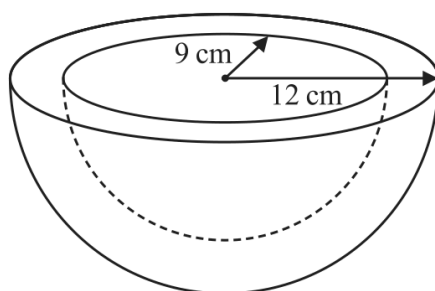


Diagram **NOT**
accurately drawn

The outer radius of the bowl is 12 cm

The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.

Give your answer correct to the nearest whole number.

..... cm³

(Total for Question 22 is 3 marks)

03 Surface Area

List every face and add their areas. Curved surfaces: $2\pi rh$, πrl , $4\pi r^2$ — then add the flat faces if the shape is solid.

WHEN YOU SEE → YOU DO

“total surface area”

→ curved parts + every flat face

a triangular prism

→ 2 triangles + 3 rectangles

“apart from the base”

→ leave that face out

“show that $A = 12x^2 + \frac{1690}{x}$ ”

→ use the volume to replace y

WORKED MODEL • NOVEMBER 2024 P2H

Right triangular prism: triangle 6, 8, 10; length 15. Total surface area?

1 two triangles: $2 \times \frac{1}{2}(6)(8) = 48$

ends

2 three rectangles: $15(6 + 8 + 10) = 360$

sides

ANSWER 408 cm^2

IDEA 03

Practise: Surface Area

My questions first, then selected practice, then real past papers.

Q24 DR ESLAM Hemisphere

CH 4 • TOPIC 7 3 marks

Find the total surface area of a solid hemisphere of radius 4 cm, in terms of π .

SELECTED PRACTICE**Q25** PRACTICE Practice • Easy

CH 4 • TOPIC 7 3 marks

4 Here is a cuboid.

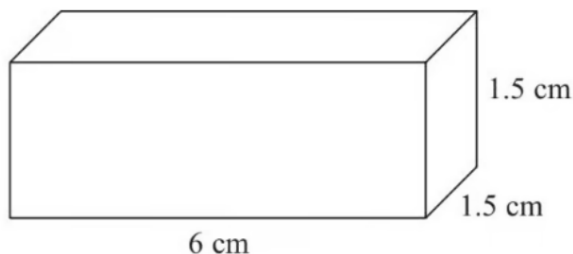


Diagram **NOT**
accurately drawn

The cuboid is 6 cm by 1.5 cm by 1.5 cm.

Work out the total surface area of the cuboid.

(3 marks)

Q26**PRACTICE**

Practice · Medium

CH 4 · TOPIC 7 5 marks

14 Jeremy has to cover 3 tanks completely with paint.

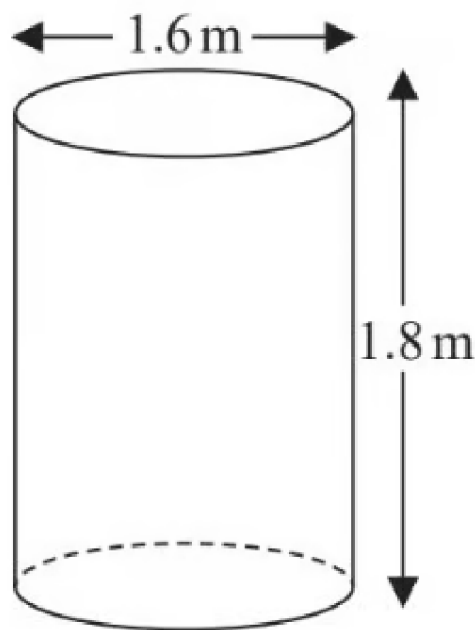
Each tank is in the shape of a cylinder with a top and a bottom.
The tank has a diameter of 1.6 m and a height of 1.8 m.

Jeremy has 7 tins of paint.

Each tin of paint covers 5 m^2

Has Jeremy got enough paint to cover completely the 3 tanks?

You must show how you get your answer.



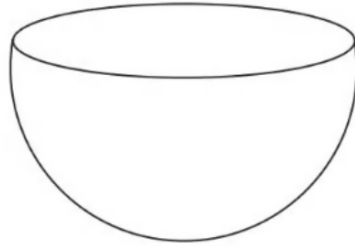
(5 marks)

Q27**PRACTICE**

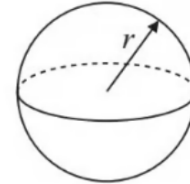
Practice · Hard

CH 4 · TOPIC 7 4 marks

- 6 The diagram shows a solid hemisphere.



Volume of sphere = $\frac{4}{3}\pi r^3$
Surface area of sphere = $4\pi r^2$



The volume of the hemisphere is $\frac{250}{3}\pi$

Work out the exact total surface area of the solid hemisphere. Give your answer as a multiple of π .

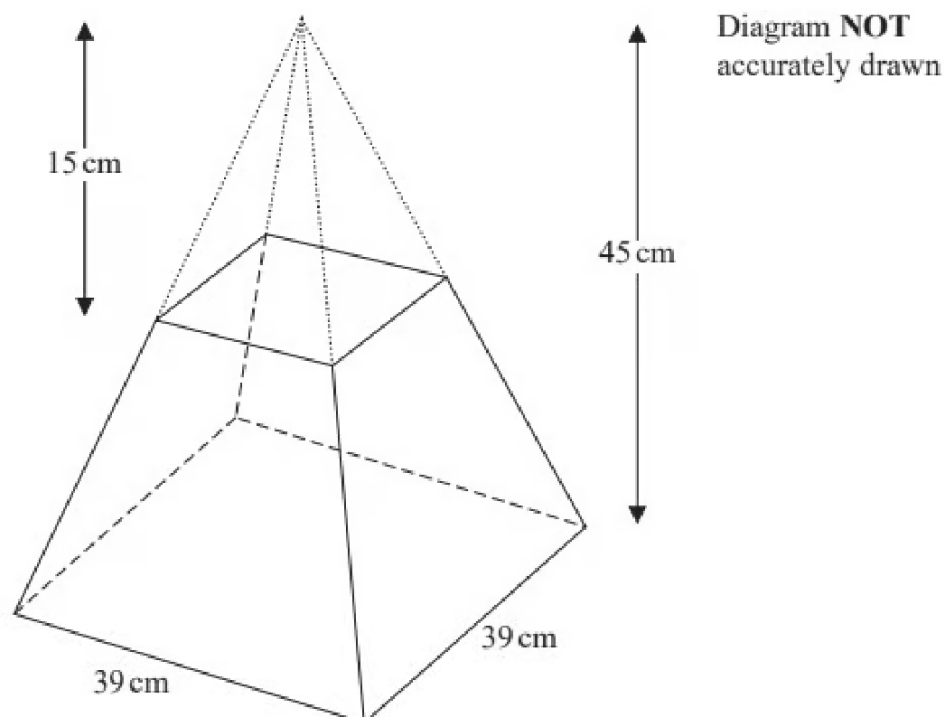
(4 marks)

Q28**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 7 5 marks

- 20 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.



The height of the small pyramid is 15cm.

The height of the large pyramid is 45cm.

The square base of the large pyramid has side length 39cm.

Work out the **total** surface area of the frustum. Give your answer correct to the nearest whole number.

(5 marks)

PAST PAPERS

Q29

PAST PAPER

Edexcel 4MA1 · January 2021 · Paper 2H · Q17

CH 4 · TOPIC 7 7 marks

17 The diagram shows a solid prism $ABCDEFGH$.

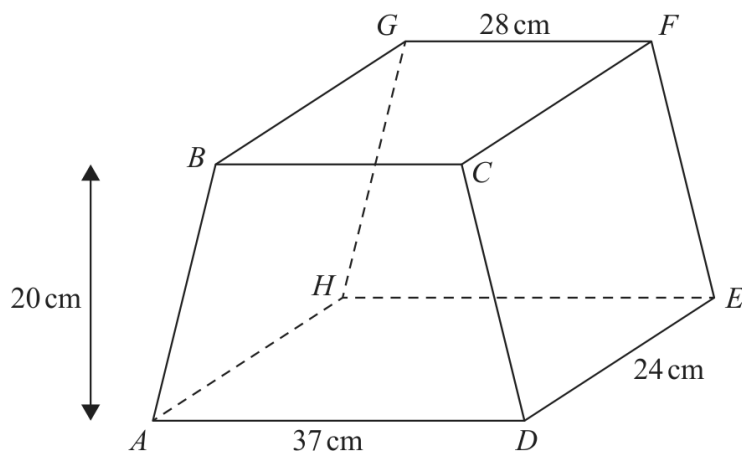


Diagram **NOT**
accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism.

The base $ADEH$ of the prism is a horizontal plane.

$ADEH$ and $BCFG$ are rectangles.

The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$$AD = 37 \text{ cm} \quad GF = 28 \text{ cm} \quad DE = 24 \text{ cm}$$

The perpendicular distance between edges AD and BC is 20 cm.

(a) Work out the total surface area of the prism.

..... cm^2
(4)

(b) Calculate the size of the angle between AF and the plane $ADEH$.

Give your answer correct to one decimal place.

.....
°

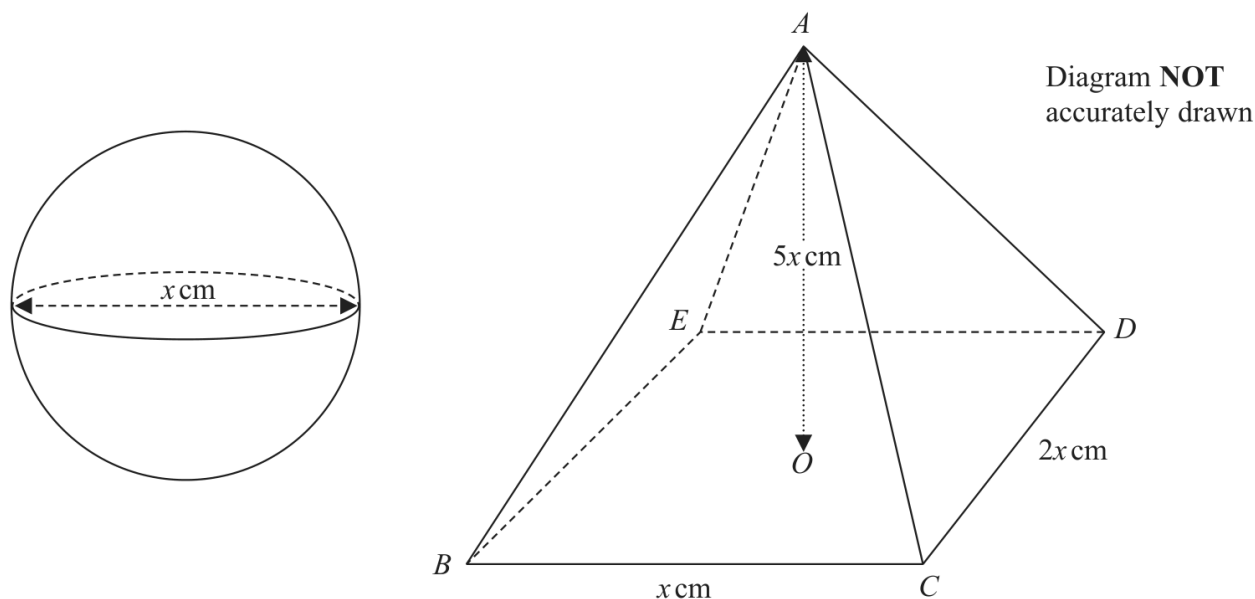
(Total for Question 17 is 7 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q22

CH 4 • TOPIC 7 6 marks

- 22 The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is 288π cm³

Calculate the total surface area of the pyramid.

Give your answer correct to the nearest cm²

..... cm²

(Total for Question 22 is 6 marks)

Q31 **PAST PAPER** Edexcel 4MA1 • November 2024 • Paper 2H • Q6

CH 4 • TOPIC 7 **3 marks**

6 The diagram shows a solid triangular prism.

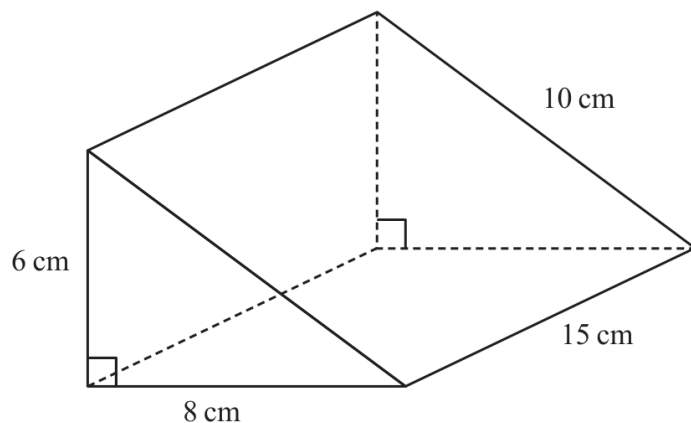


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the triangular prism.

..... cm^2

(Total for Question 6 is 3 marks)

Q32

PAST PAPER

Edexcel 4MA1 · June 2025 · Paper 2HR · Q26

CH 4 · TOPIC 7 5 marks

26 The diagram shows a solid hemisphere, H

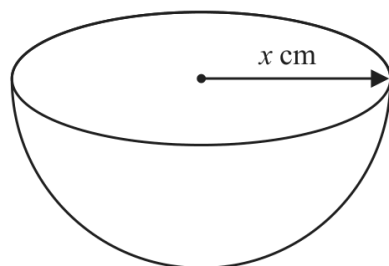


Diagram **NOT**
accurately drawn

The radius of H is x cm

The volume of H is $6174\pi \text{ cm}^3$

A bowl is made by removing a solid hemisphere from H such that the uniform thickness of the bowl is 2 cm

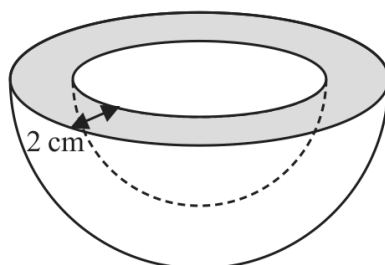


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the bowl.
Give your answer in terms of π

..... cm^2

(Total for Question 26 is 5 marks)

Q33 PAST PAPER Edexcel 4MA1 · June 2025 · Paper 2H · Q21

CH 4 · TOPIC 7 3 marks

21 The diagram shows a cuboid.

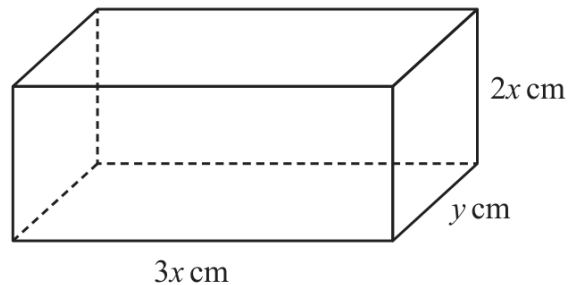


Diagram **NOT**
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A \text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

04 Problem Solving with Volumes

Rate problems: $\text{volume} \div \text{rate} = \text{time}$. Melting and recasting: the volume stays the same. Always check the units.

WHEN YOU SEE → YOU DO

“pumped at a constant rate”

→ $\text{volume removed} \div \text{time} = \text{rate}$

“melted down and made into a sphere”

→ equal volumes, solve for r

“will the cuboid be full?”

→ compare the volumes

litres and cm^3

→ $1 \text{ L} = 1000 \text{ cm}^3$; $1 \text{ m}^3 = 1000 \text{ L}$

WORKED MODEL • JUNE 2025 P1HR

A full cylinder (radius 10, height 33) is poured into a cuboid 35 by 28 by 20 with water 9 cm deep. Show it will not overflow.

1 $\pi(10)^2(33) = 10\,367 \text{ cm}^3$

water in the cylinder

2 $\text{space left} = 35 \times 28 \times 11 = 10\,780 \text{ cm}^3$

empty part

ANSWER $10\,367 < 10\,780$ — not full

IDEA 04

Practise: Problem Solving with Volumes

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q34****PRACTICE**

Practice · Easy

CH 4 · TOPIC 7 2 marks

- 10** Ashraf is going to put boxes into a crate.

The crate is a cuboid measuring 2.5 m by 2 m by 1.2 m
Each box is a cube of length 50 cm

He does these calculations.

$$\begin{aligned}\text{volume of crate} &= 2.5 \times 2 \times 1.2 \\ &= 6 \text{ m}^3 \\ \text{volume of one box} &= 0.5 \times 0.5 \times 0.5 \\ &= 0.125 \text{ m}^3 \\ \text{number of boxes} &= 6 \div 0.125 \\ &= 48\end{aligned}$$

He claims,

"I can put 48 boxes in the crate."

Evaluate Ashraf's method **and** claim.

(2 marks)

Q35**PRACTICE**

Practice • Medium

CH 4 • TOPIC 7

4 marks

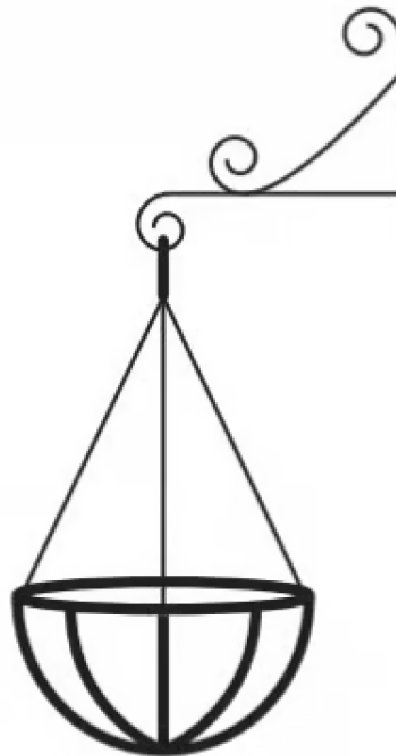
8 Anne wants to fill 12 hanging baskets with compost.

Each hanging basket is a hemisphere of diameter 40 cm.

Anne has 4 bags of compost.

There are 50 litres of compost in each bag.

Has Anne got enough compost to fill the 12 hanging baskets?



hanging basket

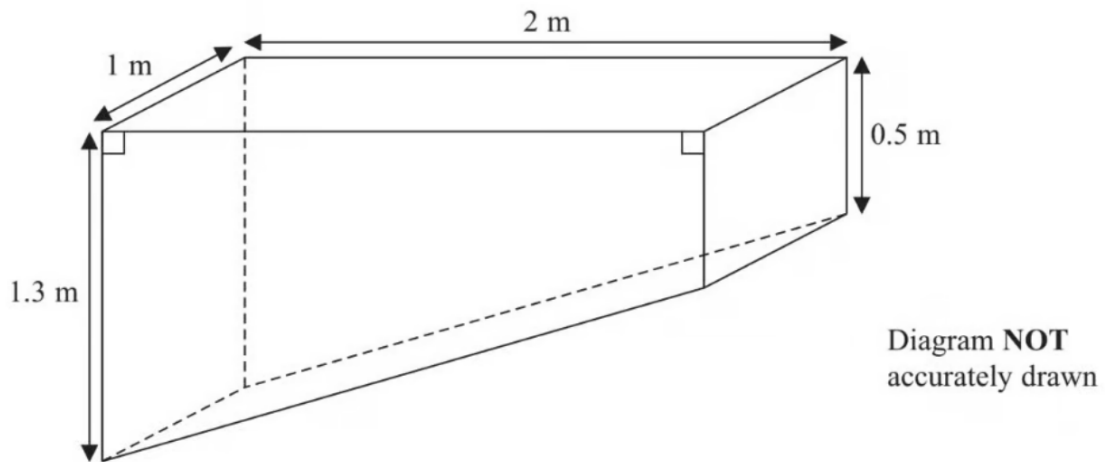
(4 marks)

Q36**PRACTICE**

Practice · Hard

CH 4 · TOPIC 7 6 marks

- 11 Sumeet has a pond in the shape of a prism.



The pond is completely full of water.
Sumeet wants to empty the pond so he can clean it.
Sumeet uses a pump to empty the pond.

The volume of water in the pond decreases at a constant rate. The level of the water in the pond goes down by 20 cm in the first 30 minutes.

Work out how much more time Sumeet has to wait for the pump to empty the pond completely.

(6 marks)

Q37**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 7 4 marks

15 A ball contains 5000 cm^3 of air.

More air is pumped into the ball at a rate of 160 cm^3 per second. The ball is full of air when it becomes a sphere with radius 15 cm



Does it take less than 1 minute to fill the ball?
You must show your working.

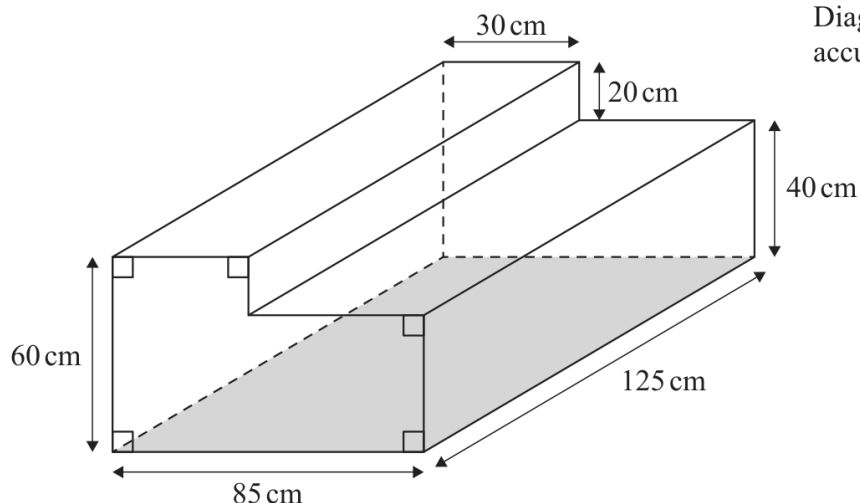
(4 marks)

PAST PAPERS

Q38 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 1H · Q3

CH 4 · TOPIC 7 4 marks

- 3 The diagram shows a container for water in the shape of a prism.



The rectangular base of the prism, shown shaded in the diagram, is horizontal.
The container is completely full of water.

Tuah is going to use a pump to empty the water from the container so that the volume of water in the container decreases at a constant rate.

The pump starts to empty water from the container at 10 30 and at 12 00 the water level in the container has dropped by 20 cm.

Find the time at which all the water has been pumped out of the container.

(Total for Question 3 is 4 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q10

CH 4 • TOPIC 7 6 marks

10 Jonty has a storage container in the shape of a cuboid, as shown in the diagram.

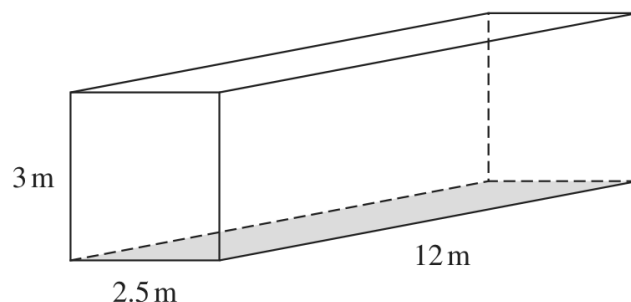


Diagram **NOT**
accurately drawn

Jonty is going to paint the outside of his storage container, apart from the base which is shown shaded in the diagram.

He needs enough paint to cover the four sides and the top.

Each tin of paint covers an area of 15 m^2

The cost of each tin of paint recently increased by 10%

After the increase, the cost of each tin of paint is £26.95

Jonty says

“**Before** the increase, I could have bought enough paint for less than £200”

Show that Jonty is correct.

Show your working clearly.

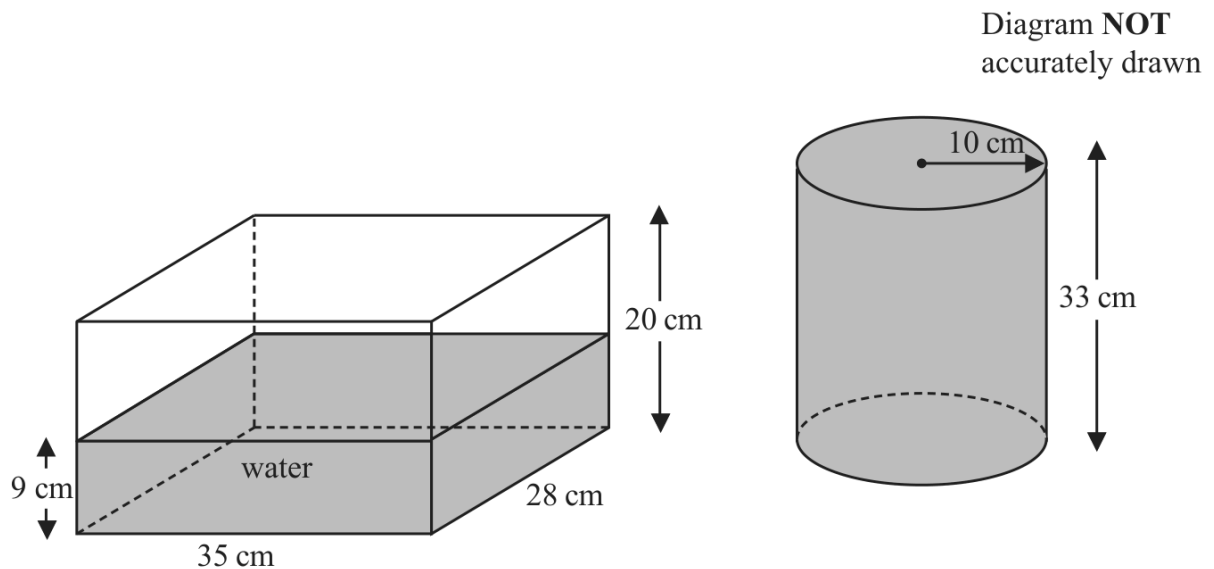
(Total for Question 10 is 6 marks)

Q40**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q10

CH 4 · TOPIC 7 3 marks

- 10 The diagram shows two water containers.
One is a cuboid and one is a cylinder.



The cuboid measures 35 cm by 28 cm by 20 cm

The surface of the water in the cuboid is 9 cm above the base of the cuboid.

The cylinder has a radius of 10 cm and a height of 33 cm

The cylinder is completely full of water.

Izzy is going to pour all the water from the cylinder into the cuboid.

Show that the cuboid will not be completely full of water.

(Total for Question 10 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|---|
| 01 | Prisms | Cross-section $\times$ length; cylinder $\pi r^2 h$. |
| 02 | Pointed / round | $\frac{1}{3}$ for cones and pyramids; $\frac{4}{3}\pi r^3$; frustum = big – small. |
| 03 | Surface | Every face; $2\pi rh$, πrl , $4\pi r^2$. |
| 04 | Problems | Rate = volume $\div$ time; recast: same volume. |

BOTTOM LINE Name the solid $\rightarrow$ formula $\rightarrow$ units (cm^3 , litres, m^3).

Answers

Volume and Surface Area • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • PRISMS AND CYLINDERS

Q1	Dr Eslam • Prism	$18 \times 12 = 216 \text{ cm}^3$
Q2	Dr Eslam • Cylinder	$\pi(25)(8) = 200\pi \text{ cm}^3$
Q3	Practice • Easy	cross-section 38 cm^2 : 380 cm^3
Q4	Practice • Easy	$h = \frac{3.8}{0.25\pi} = 4.84 \text{ m}$
Q5	Practice • Medium	$x = 12$
Q6	Practice • Very Hard	$AC = 13, CM = \sqrt{87}: CF = 18.65: 560 \text{ cm}^3$
Q7	Edexcel 4MA1 • January 2020 • Paper 2HR • Q13	$h = 6.4$
Q8	Edexcel 4MA1 • January 2021 • Paper 1HR • Q5	(a) $x = 4.5: 573 \text{ cm}^3$ (b) $1\,000\,000 \text{ cm}^3$
Q9	Edexcel 4MA1 • May 2021 • Paper 2H • Q10	467 cm^3
Q10	Edexcel 4MA1 • June 2023 • Paper 2HR • Q9	3790 cm^3
Q11	Edexcel 4MA1 • November 2023 • Paper 1H • Q21	7638 cm^3
Q12	Edexcel 4MA1 • November 2024 • Paper 1H • Q5	277 litres
Q13	Edexcel 4MA1 • June 2025 • Paper 1H • Q7	$h = 19.4$

IDEA 02 · CONES, PYRAMIDS AND SPHERES

- Q14** Dr Eslam · Cone $\frac{1}{3}\pi(36)(10) = 377 \text{ cm}^3$
- Q15** Practice · Medium $\frac{128}{3}\pi + \frac{160}{3}\pi = 96\pi = 302 \text{ cm}^3$
- Q16** Practice · Hard $r = 14.8 \text{ cm}$
- Q17** Practice · Very Hard $\frac{2}{5}h \text{ cm}$
- Q18** Practice · Very Hard $x = \frac{12}{5}$
- Q19** Edexcel 4MA1 · January 2020 · Paper 2H · Q26 $x = 107$
- Q20** Edexcel 4MA1 · June 2022 · Paper 1H · Q22 $k = \frac{16000}{3} + \frac{8400}{3} = 8133\frac{1}{3}$
- Q21** Edexcel 4MA1 · June 2023 · Paper 2HR · Q24 $r = 1.5 \text{ cm}$
- Q22** Edexcel 4MA1 · May 2024 · Paper 1H · Q23 1090 cm^2
- Q23** Edexcel 4MA1 · November 2024 · Paper 1H · Q22 $666\pi = 2092 \text{ cm}^3$

IDEA 03 · SURFACE AREA

- Q24** Dr Eslam • Hemisphere $2\pi(16) + \pi(16) = 48\pi \text{ cm}^2$
- Q25** Practice • Easy 40.5 cm^2
- Q26** Practice • Medium 39.2 m^2 needed, 35 m^2 of paint: No
- Q27** Practice • Hard $r = 5: 75\pi \text{ cm}^2$
- Q28** Practice • Very Hard 5090 cm^2
- Q29** Edexcel 4MA1 • January 2021 • Paper 2H • Q17 (a) 3844 cm^2 (b) 26.3°
- Q30** Edexcel 4MA1 • November 2021 • Paper 2H • Q22 $x = 12: 2469 \text{ cm}^2$
- Q31** Edexcel 4MA1 • November 2024 • Paper 2H • Q6 408 cm^2
- Q32** Edexcel 4MA1 • June 2025 • Paper 2HR • Q26 $x = 21: 1684\pi \text{ cm}^2$
- Q33** Edexcel 4MA1 • June 2025 • Paper 2H • Q21 $y = \frac{169}{x^2}, A = 12x^2 + \frac{1690}{x}$

IDEA 04 • PROBLEM SOLVING WITH VOLUMES

- Q34** Practice · Easy
volume method is wrong: only $5 \times 4 \times 2 = 40$ boxes fit
- Q35** Practice · Medium
 $12 \times \frac{2}{3}\pi(20)^3 = 201 \text{ L} > 200$: No
- Q36** Practice · Hard
 1.8 m^3 ; 0.4 m^3 in 30 min: 105 more minutes
- Q37** Practice · Very Hard
 $\frac{4}{3}\pi(15)^3 - 5000 = 9137$: $57.1 \text{ s} < 60$: yes
- Q38** Edexcel 4MA1 · January 2021 · Paper 1H · Q3
full $500\,000 \text{ cm}^3$; top 20 cm = 75 000 in 90 min: 600 min in total: 20:30
- Q39** Edexcel 4MA1 · January 2022 · Paper 1HR · Q10
 117 m^2 : 8 tins at £24.50 = £196 < £200
- Q40** Edexcel 4MA1 · June 2025 · Paper 1HR · Q10
 $1000\pi \times 33 = 10\,367 \text{ cm}^3 < \text{space } 35 \times 28 \times 11 = 10\,780$